Achieving Creativity With Artificial Neural Networks

One of the recent applications of neural networks has been in the field of art and music. Can creativity be accomplished through artificial neural networks (ANN)? ANNs are capable of some level of creativity, although not on par with the human faculty. What are the challenges of achieving music and art through algorithms? What are the various approaches? These are some of the questions we shall talk about in this article

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Over decades, ANNs have come a long way from single layer perceptrons that solved simple classification problems to highly complex networks for deep learning. Today, neural networks are used for more than just classification or optimisation problems in various fields of application. One of the most recent applications of ANNs is in the field of art and music. The goal of such an application is to simulate human creativity and to serve as an assisting tool for artists in their creations.

Early works began as early as 1988

One of the first works of using ANNs for music composition was carried out as early as 1988 by Lewis and Todd. Lewis used a multi-layer perceptron, while Todd experimented with a Jordon auto-regressive neural network (RNN) to generate music sequentially.

The music composed with RNNs algorithmically had global coherence and structure issues. To circumvent that, Eck and Scmidhuber used long short-term memory (LSTM) neural networks to capture the temporal dependencies in a composition.

In 2009, when deep learning networks became more interesting for research, Lee and Andrew Ng started using deep convoluted neural networks (CNN) for music genre classification. This formed a basis for advanced models that used high-level (semantic) concepts from music spectrograms.

Most recently, wavenet models and generative adversarial networks (GAN) have been used for generating music. Waveform based models outperformed spectrogram based ones, provided enough training material was available.

A multitude of network models

In the following section, we shall discuss the various ANN models that are used to create music and art.